The Social Radars

S2: Trevor Blackwell, Co-Founder Y Combinator; Founder, Anybots

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Jessica: I'm Jessica Livingston, and Carolynn Levy and I are the Social Radars. In this podcast, we talk to some of the most successful founders in Silicon Valley about how they did it. Carolynn and I have been working together to help thousands of startups at Y Combinator for almost 20 years. Come be a fly on the wall as we talk to founders and learn their true stories.

Carolynn, I am so excited because today we have on the show my friend and Y Combinator co-founder Trevor Blackwell.

Carolynn: Welcome Trevor.

Jessica: Welcome Trevor.

Trevor: Thanks.

Jessica: Glad you could be here.

Trevor: Yeah, no, I've been wanting to do your podcast since you launched it.

Carolynn: We've already heard a lot about you from Paul Graham.

Jessica: I know. Your name crops up a bit here and there, and so ... Well, we'll definitely have some specific questions that we want to clarify from the Paul Graham interview, but I want to go back a little bit to how did you make it from where you grew up in rural Canada to Harvard's PhD program?

Trevor: Well, I grew up in Saskatoon, Saskatchewan, north of North Dakota for people with more American geography references. It was a small city and there were only a handful of people that were into computers when I was there in high school, and I'd read about Silicon Valley. I mean, I had an Apple. My mom somehow managed to get me an Apple II computer when I was a kid, and I read everything about it and about how Jobs and Wozniak had created this thing in their garage. And I imagined them with their fingers freezing in the garage because that's what Saskatchewan garages are like.

Jessica: Even though they were in California. You were in Saskatchewan.

Trevor: Well, I'd never been to California

Carolynn: Sunny Sunnyvale.

Trevor: So I always wanted to go there to see it, but I ended up in Ottawa for college at Carleton University, studied engineering and computer science there. And then I started working for Bell Northern Research, which was kind of like Bell Labs for Canada. It was the research arm of the Canadian phone company.

Jessica: After college?

Trevor: Well, actually during college in the summers and even in the evenings after college. And they were doing a joint project with the guy who ended up being my advisor at Harvard, H.T. Kung. And so I got to know him through that project and they put in a good word for me, and I got accepted there. So I put everything in the car and moved to Boston and started at Harvard.

Jessica: So what did it feel like to show up in bustling Cambridge, Massachusetts from Saskatoon?

Trevor: Yeah, it was an eye-opener. All kinds of things I'd never really seen before. But I got there and it was this transition to suddenly being in a place where almost everyone is smarter and more knowledgeable about the subject than I am. So that lit a fire under me. Had to try and catch up.

Jessica: But you must have caught up pretty soon because you met Robert Morris, because you were getting your PhDs at the same time, right?

Trevor: Yeah, we had the same advisor and our offices weren't that far from each other.

Jessica: So Paul had started Viaweb and he was good friends with Robert, and he asked Robert, who's the best programmer in your program? And he said you. So you must have caught up very quickly.

Trevor: I guess so.

Jessica: Very modest. You're very modest, Trevor.

Trevor: I mean, I was good at the practical stuff, and it was the theory that I felt insecure about because I'd studied computer engineering rather than computer science. So I could program, but they would keep talking about these complicated theory things that took me a while to catch up on.

Jessica: Well, Paul described you as basically a programming monster doing prodigious feats of programming, which now that I know you, I believe. But do you remember when you were with Robert in the program and he said, "Hey, I'm working on this startup."? Do you remember getting introduced to Paul and the whole Viaweb thing?

Trevor: Yeah. He introduced me to Paul and Paul explained this idea that lots of people would want to sell things on the internet, which I believed because this was late '94 and Amazon sort of existed. There were a couple of other bookstores on the web. I just loved it because I could order any book I wanted if I just knew the title instead of whatever might be available at the campus bookstore. So I liked the idea of there being more things for sale on the internet, and I believed the idea that every catalog company would eventually need to be on the internet, which was a bold idea in 1994.

Jessica: And you probably dealt with your fair share of catalogs being an electrical engineer, right? Didn't you have huge catalogs?

Trevor: Well, yeah. I mean, in Saskatoon there was not a retail store for computers or electronics. I'd order stuff from ... DigiKey existed then in the '80s. That's still the standard place to buy electronics components. But at the time, it was in the back of a magazine, there'd be this page with a form to fill out for quantity and price and whatever, and you'd send it in with my dad's credit card number. And so being able to order stuff on the internet seemed great.

Jessica: I'm dying to know what your first meeting with Paul was like. What did you think of him? What was he like? What stage was Viaweb at?

Trevor: Well, what actually got me interested in it was that it was a software business that didn't require writing Windows software and putting it in a box and shipping it with version numbers. So kids these days don't know how horrible that was, but in the pre-internet days of software, to sell a piece of software, you would write it and then you would burn it on CDs, put it in a box with a nice glossy cover on it and sell it. And if there was something wrong with it, if there was a bug in it, too bad. The users had to suffer with it until next year when the next version came out. Because there were no updates, no downloads.

So that's a terrible way to write software because knowing that any mistake is going to take a year to fix, you probably want to spend six months testing every aspect of it. You've got to test it on people's weirdly configured computers and with other software, and it was such a grind to just make a simple piece of software that did something. So not many people wanted to be in that business and certainly not me. So when Paul pitched this idea, we could write all the software on our Unix systems that we liked from using them in grad school and run it all on our own hardware and update it whenever we liked. To me, that's what got me excited about it.

Jessica: Okay. The novelty of the web-based software, basically.

Carolynn: You'd never heard of anyone doing this before? This was not a thing you'd ever heard of?

Trevor: No, not really. It existed for open source software, but I can't really think of someone who made a business doing software in that way.

Jessica: So how did they reel you in? How did they entice you to get involved? Because you were getting your PhD.

Trevor: Yeah, no, I had official things to work on. Well, Paul showed me this demo of this thing that I'd never seen before, which was using a web browser, but when you clicked on a link, instead of just going to a page that link pointed to, it would do something, it would change something on the system. So he had this little demo. It had no colors or anything, it was just blue links on a gray background. But you could create data on the server just by clicking links. And I thought, "Wow. This is great. Someday all software is going to work like this." And so I wanted to just try writing software that way to see if it really was possible. So I built this completely separate thing that just did that. I started it on my birthday, which is why I remember the date. I thought, it's my birthday, I can do whatever I want. I was in school, and didn't want to work on my PhD, so I thought I'll just write one of these things where you can click on links and it does stuff.

Jessica: Did you write it in Smalltalk Trevor?

Trevor: Yeah, I wrote it in Smalltalk, which is ... It was a good language to do it in. Lisp was also a good language to do it in. Yeah, I wrote it in Smalltalk and then a couple days later I showed that to Paul and said, "Hey, check this out." And so he could see that I'd gotten the idea and then we started working on it. It was quite a while before we got official about stock or paychecks or anything like that, but we decided I would be part of this thing.

Carolynn: Did you end up finishing your PhD, or what was the timeline in terms of you getting really involved in Viaweb or in graduating? What happened there?

Trevor: They overlapped in the most difficult and stressful way. I did finish my PhD. I graduated on the same day we signed the acquisition deal with Yahoo.

Jessica: Don't you remember that, Carolynn, from Paul's interview?

Carolynn: I do now. Yeah, yeah. Okay. Yeah.

Jessica: Insane.

Trevor: Yeah, I had to duck out of the ceremony in my crimson robes running through Harvard Square to sign this thing.

Jessica: No way. You literally ducked out of your graduation?

Trevor: Yeah. But months before that I realized actually I really like building this thing. I want to keep working on this company. I think it's going to be great. But I said I'd do a PhD, so I got to write up a thesis. So yeah, I took a bunch of the things I'd been working on and put them all into one big file and started editing them until it kind of made sense. And that was-

Jessica: For your dissertation, you mean?

Trevor: Yeah, yeah.

Jessica: Looking back on your dissertation, does it feel like it was a fascinating one or did you just say, I have to write a dissertation, I'm working on this kind of thing, let's make this work.

Trevor: I think there's some value in it. There's one core idea, and then there's a couple of weakly related examples of how to use it. Those are just the things I'd been working on, and I had this idea for a while and thought this would be a good way of ... It's a way of measuring the performance of systems. When you're trying to make a computer go faster or some software go faster, you got to measure it. And there are lots of ways of getting fooled because computers just go faster and slower sometimes for mysterious reasons. And so I had this way of making more robust performance measurements in the back of my head for a few years and then just tinkering around with things I had used it on a couple of projects. And so I thought, okay, that's going to be my thesis. I got two chapters already.

Jessica: Does this have anything to do with the Bogometer or nothing to do with the Bogometer?

Trevor: No, not really. Maybe the name Bogometer comes from that, but ... Well, okay, so the Bogometer ... This was another new idea that no software company had probably ever had this opportunity before, but we had software that was running on our servers and the disks were clicking away, and you could notice in the middle of the day the disks would be clicking away more like disks used to click back then.

Carolynn: I remember.

Trevor: So we thought, let's make a big mechanical dial that has a clock hand that goes back and forth that shows how many hits per second and how many orders we're getting. I mean, at first we were getting zero orders per hour. Maybe one. So that dial didn't have much range. So I made this thing with one dial that was the number of hits per second, and so I had a motor and a big pointer on it, and I could laser print a scale so I could change it when we started growing. And so I put all these things together and connected it to a serial port on one of the computers, and it showed us how many hits per second we were going. And it ended up being always one of the little highlights of the tour. We'd have investors or potential employees or something or customers coming and we'd say, "Look, we're getting 4.5 hits per second on our servers."

Carolynn: But why is it called the Bogometer? What's the Bo?

Trevor: Paul suggested this idea, and I thought, oh, that's so bogus. It doesn't matter how many hits we're getting this second as long as we know day to day. So I called it the Bogometer, and then it actually ended up being a really good idea. Because sometimes something would go wrong, the internet would go down or something and all the bells would go ... And you could hear them because they made a little whizzing sound, so it was-

Jessica: So you'd say, "Whoa. Something's going wrong." Now, was the Bogometer in your office, which actually was also the server room?

Trevor: Well, maybe one version was inside, but I put it on the wall outside so people would look in and there'd be this rack of slightly wobbly servers, but then this ... Because it grew to have I think eight dials on it showing all kinds of things like revenue and-

Carolynn: Oh, let me pause you on that theme because I know we're going to talk about robots soon, and I'm just wondering, building knobs and dials, is this very early robot stuff or had you already been building robot stuff as a hobby on the side?

Trevor: Oh, I started building robots in middle school. Didn't do a lot for my social popularity, but it was fun. And they were pretty clunky early on. The first one was a big old oil drum, one of those green and white striped drums. And I put some holes in the bottom and had these Army surplus motors and a box with switches on it to control it, and I eventually put an arm on it. So I'd been building robots. That's what I always wanted to do, I think.

Carolynn: Okay. So the Bogometer was not your debut robotic thing? This is a hobby that goes way back. Okay.

Trevor: No. No.

Jessica: One of Trevor's many amazing side projects that probably could be a standalone product at a company, but he just whips it up on the side.

Trevor: It would be a good startup. I keep hoping one of the metrics as a service companies will have that as an option. Like a wall display for your office with real-time stats.

Jessica: Do they not have that?

Carolynn: Yeah, I would think someone would.

Trevor: I haven't seen it.

Carolynn: Who goes into the office anymore?

Jessica: All right. You heard it here first. Great new idea. Trevor, do you remember when the power went out in Cambridge and you guys had to keep the servers ... Again, I'm going to reiterate, they were in your office. You had an office in the server room.

Trevor: They were tower servers sitting on one of those wire racks, so they kind of wobbled a lot whenever someone walked by.

Carolynn: Wasn't it hot? Those rooms get hot, right? Wouldn't it be hot in there?

Trevor: They did, yeah. No, they were quite hot. With one, it wasn't so bad, but at some point we had maybe six of them. So then I got a window air conditioner and that wasn't enough so I ended up with stacked four window air conditioners to try to keep this room, which was like ... We were right in Harvard Square, which is a great location because it was just a few minutes walk from my grad school office, and there were restaurants and shops. But this thing had been a house, I guess a long time ago, and someone converted the third floor to be an office, and it had been a professional office.

Jessica: It was a wooden house, right?

Trevor: Yeah, old wooden, probably 1900 or so structure. And it had a bar in the bottom, this Irish bar.

Jessica: That's fun.

Trevor: It got a little loud at night.

Jessica: Wasn't your office also called the hot tub because it got so hot?

Trevor: Hot tub. Yeah. Because it was hot.

Jessica: Okay, so you're in the hot tub-

Trevor: With all the servers clicking away and the fans blasting.

Carolynn: The Bogometer's going back and forth.

Jessica: So then there's a massive power outage in Cambridge, and tell me what happened relating to the generator.

Trevor: Yeah, it was a big power outage because some transformer blew up underground and they said it would be maybe a day or more to fix it. We had a battery backup system. That was surprisingly legit of us to have that, but it was good for half an hour. And so in that half an hour I thought, okay, I'm going to drive out to Costco, buy a generator and get back here in half an hour. Costco is actually pretty far. So yeah, so I zoomed over to Costco, got this horrible gas generator generator, Generac or something, maybe 10 horsepower, and lugged it up the stairs and put it out on the fire escape.

Carolynn: I was going to say, wood building, gas power generator. This just sounds like a disaster in the making.

Trevor: Yeah. Yeah, I'd filled it up at the gas station just like you pour gas in it. I didn't have a can or anything. Somehow managed. We managed to cut over to the generator and get it going, but it made such a racket that the people downstairs, like this lady downstairs who was our ... I don't know if she was exactly our landlord, but she was the landlord's agent anyway. Came and told us like, "No, no, no. You can't have that thing up here."

Carolynn: They are extremely loud, those gas power generators. Extremely.

Trevor: They are. Yeah. Especially it's a power failure in the '90s, and the city is suddenly very quiet and you can see the stars and all of a sudden ...

Jessica: What did you do?

Carolynn: Well, also if you didn't have a gas can, how were you proposed to refill it?

Trevor: One hour at a time, I guess.

Jessica: You moved the generator, did you not?

Trevor: Yeah, we moved the generator downstairs, I think, and at some point we hadn't had it on long enough to recharge the batteries and we lost power. But then eventually a couple hours later, power came back on and it was okay. But at that time we could see on the Bogometer how much revenue per hour and it was real numbers. It was tens of thousands of dollars an hour. And our merchants were losing that if our servers were down. And they would call us and yell at us, suitably angry for how much money they were losing. The upside of this kind of software is you didn't have to have these boxes with yearly releases of the new version, but the downside is something goes wrong and you've got to get up in the middle of the night and fix it.

I wore the pager for basically the whole history of Viaweb. There were lots of things that went wrong so a lot of late nights.

Jessica: Late nights in the server room?

Trevor: Yeah.

Jessica: But the generator ... I mean, what I remember from this story, and that's why I wanted you to tell it, wasn't it on the sidewalk with 20 different extension cords going all the way up to the third floor into the window through the office to the other side of the back of the office where the servers were tight as a pin, so tight and chest level, there was no slack in the extension cord.

Sorry. I'm sure the story won't seem that good on the podcast, but it's one of my favorites. I just love it. Were there any electrical engineering adventures when you were at Viaweb? Because Paul said you used to do a lot of crazy things in the electrical engineering department.

Trevor: A little bit. A lot of our users wanted their orders faxed because the people in the warehouse probably didn't have email. So someone would place an order online and our system would fax it to a fax machine in the shipping warehouse. And at first we just had one fax line, but that soon became not nearly enough and so we had this big rack of fax modems, 15 or more, all wired up to the phone closet. We were pushing the limits of how many phone lines you could get into one of these old three story walk-ups.

So there was that. Various adventures in trying to keep the servers up all the time. We started off, we had one server, and if it was down, nobody even noticed before we had many users. And then just gradually, gradually it started getting more critical that we couldn't have the servers be down. So at some point we thought we should get a battery backup. Because if the power fails, it's not just that it's down, but at least at that time, shutting those kinds of computers down without running a shutdown command could corrupt the disk and then you could be in big trouble. So we got this battery backup thing, but the computers were already plugged into the wall and we wanted to plug them into the battery backup system, so there's no proper way of doing it. But I made this cord that had a mail plug on both ends so I could bridge between the power strips that were running the servers and the output of this UPS battery backup system. Surprisingly, it did not blow up.

Jessica: Yeah, I was going to say that sounds really scary. This does sound scary. Oh my God. Did you say, "Oh, okay, it should work now."? For the listeners, that's like the famous Trevor line. Oh, okay. It should work now in his Canadian accent. And would you say 50% of the time it would work?

Trevor: Yeah.

Carolynn: And the thing is you guys couldn't move because you're all still at Harvard, so you can't move to a more convenient location to put your booming business into.

Trevor: We went around and looked. We looked at legit-looking offices like glass towers on Route 128 and they were so soulless and grim.

Jessica: Yeah.

Trevor: I probably could have put up with it, but Paul's pretty sensitive to-

Jessica: Environment. Yeah.

Trevor: The atmosphere and he just wasn't going to do it.

Jessica: He had his routine and he probably ... I mean, I didn't know Paul during Viaweb, but I know for sure he had a routine where he probably got his breakfast, went into program and wanted to walk to and from work, did not want to have to commute out to Route 95. What was Paul like to work with back then?

Trevor: I think the defining dynamic was that he would insist that there was a right way of doing something, and I'd be like, "Oh, come on. No. That can't be the right way." And then I'd go around and think about it for a while and then actually realize he was right. And if that only took an hour, it was no problem. But sometimes I'd build the other version of it and show it to him, and it was still wrong, and eventually I had to realize, yeah, actually he was right.

Jessica: That sounds familiar. Sounds familiar. Actually, I have to share with you, Paul described you this morning as an extremely powerful engine with a rather small rudder. I love that. And knowing you, I just love that. Oh my gosh.

Okay. So Yahoo, as we heard, you sign the deal with Yahoo to get acquired on the day you got your PhD. You move out then to Mountain View.

Trevor: Yeah, so it was clear this was going to be the next chapter of our lives in California so we moved out there, bought a place. We had to move a dozen employees too. Had to start using Yahoo's server infrastructure instead of our racks in the walk-up. And so yeah, it was a very busy summer.

Carolynn: Were you excited to move to California?

Trevor: I was, yeah. No. As soon as I came out and we were driving around and there's Apple and there's Hewlett Packard and Silicon Graphics and all these great companies whose stuff I've been using for years and years and just like there they are, and the employees are walking in and out.

Jessica: And let's not forget though, that Yahoo back then in 1998 was like ground zero of the internet bubble, right?

Trevor: Yeah. No, they were hot. Well, they got a lot hotter after they bought us. Not just because of us, but maybe we helped a little. But yeah, they were the darling of the internet.

Jessica: So what was it like working there? Did it feel like a big company yet, or did it still feel startup-y?

Trevor: Well, at first it felt really quite startup-y. It was a small number of people just doing the work. They had crappy offices in a two-story industrial park in Santa Clara. And they basically left us alone to just ... We had a business that was working, the technology worked. They just left us alone to do that. But we had more resources. We could get hardware and much more internet bandwidth. And then they started asking us to integrate with some of their systems, which made sense using the Yahoo login and stuff like that. And that was all good. But I mean, it definitely turned into a big company pretty quickly. I think they felt that they were supposed to be a big company, and so they had to start doing all these big company things even though no one there really had much experience with doing big company things in a way that worked. So one of the things that was frustrating was at Viaweb, we just talked to our users. We knew dozens of them by name, and we could just email them or call them up and ask them questions about what should the software do if this happens.

But Yahoo had this group of user interaction research, and so they wanted to do all the user interviews. And they would produce these nice reports with statistically valid observations from users, and we'd get them two or three months after we had asked the question, which by then I'd probably already completely changed what the software did already. So yeah, that worked poorly. I mean, maybe it worked okay for Yahoo's core business of search and internet directory, but it didn't really work very well for us.

Carolynn: What happens when you give them that feedback? Are they like, "Oh yeah, thanks," and they just keep doing their same thing?

Trevor: Well, I could have been a better corporate person. I would just get frustrated and do it my own way rather than try to change the system. I did once try to change something big there, which was at some point I thought basically the core product that Viaweb became, which became Yahoo Store, I thought, it feels like it's 90% done. I'm mostly doing maintenance. I want to try something new. And so my idea was that we had all these merchants who we knew would want to buy ads online for their stuff pointing to their store, which wasn't really a thing then. At the time, the banner ads were for TV channels or gasoline stations, or the same thing you would see on TV. It was like TV ads or magazine ads. But I figured most of our merchants would spend $1,000 a month on advertising their store and driving traffic to it. So I wanted to make that advertising product. Google later succeeded with AdWords where you can just buy a few thousand dollars of ads with a credit card. But at the time, in order to buy ads, you had to go talk to an ad salesman and they weren't interested in talking with you unless you were making at least a $100,000 ad buy. So there's no way you could just buy a few ads with a credit card.

Jessica: So did you build something?

Trevor: I did. I built it and it worked great. So I built this prototype and went and showed it to the ad sales people and was like, "Look, we can now sell ads to this whole new category of users." And they didn't get it. It was a case study in crossing the chasm kind of thing where they had this existing business of selling multi-million dollar ad buys to big companies, to Procter and Gamble and to advertising toothpaste and whatnot. And so this idea of you're going to sell $1,000 of ads, and it would be like, no, but we'll do that a thousand times and it'll be real money. But they didn't quite get it.

But the big problem was I had to show a price for the ads to buy them. And I mean, this is a holdover from legacy media that there's a list price, which is 10 times higher than anyone ever pays. It's like the rate on the inside of the hotel room door. It cannot exceed that rate. And so they put this ridiculous rate up there, and then everyone gets charged less. So the official rate for the ads would be like $80 per thousand ads per CPM, but the actual price was like $8. That's what any of these big companies would get a deal for. And so they wouldn't let me put $8, I had to put $80.

So that kind of killed it. At the time, it would've been a tiny fraction of their revenue so it was hard to get anyone excited enough about this to change this pretty big policy and reveal what the real price might be. Because a few suckers did pay the high price.

Jessica: Oh, interesting. I have on my list of questions, why didn't Yahoo become Google?

Trevor: Oh, man. They could have.

Jessica: I think that sort of explains partly.

Trevor: Well, they should have bought Google. They would've been able to buy Google for under a billion dollars at one point, and that would've been a great deal. But yeah, it didn't happen.

Jessica: You stayed at Yahoo longer than a lot of people from Viaweb.

Trevor: Right. I stayed about three and a half years. At first, partly out of ... Well, what turned me off was this frustration with not being able to launch a new product. Because at Viaweb, Paul and I would've talked about it over lunch, and we would've launched it in the afternoon. That's the way we did things. And then doing it there, I spent months trying to talk to the right people and get people on board, and eventually it didn't even work. So what a frustrating thing that was.

Jessica: Did you try other projects or just that one?

Trevor: No, that was the main one that turned me off working for a big company.

Jessica: Then you decide to go and start Anybots. I just want the listeners to know Trevor was responsible for a really big scientific achievement, which was making the first dynamically balancing biped robot. Tell us what the dynamically balancing biped robot is and how it all got started.

Trevor: Well, after I decided I didn't want to work for a big company, I thought, okay, well, what do I want to do? I want to build something. And so I had made a list of all the things I thought that science fiction had been promising us for a long time and still didn't exist, and that I could contribute to. It didn't need new physics or anything. And robots seemed like the obvious answer. So I left Yahoo on a Friday and started a robot company the next Monday. In retrospect, I encourage people to take a break between things. It was fine because I was all energized by this exciting new idea, but it had been a long time since I'd had a break from working on Viaweb and PhD simultaneously and then trying to transition to Yahoo and then-

Carolynn: Yeah, it's a long grind.

Trevor: Yeah. So I wanted to build a walking robot. Two-legged, human sized walking robot. Because they have them in science fiction, and we don't have them in real life so something needs fixing there.

Jessica: And pick up where you left off in middle school.

Carolynn: Oil can robots.

Trevor: Yeah, basically.

Carolynn: Did you immediately rent the space at 320 Pioneer at that point, or did it take a little while before you got to that?

Trevor: I started in the basement. I bought a bunch of machine tools and aluminum and made prototype legs all in my basement, and at some point I needed a little more space so I sublet a place from Tellme, if you remember them. They did voice recognition. And so I got a huge amount of space cheap because it was the dark days of late 2001. And yeah, so I started trying to get this robot to walk. And so the first version I had the hardware. It was two legs with a hip and a knee and an ankle each. And so the very first thing I got working was I built myself a little miniature version, sort of a hand finger-sized version of this robot with sensors on it. And I had the big robot sitting over there and I could move the legs in the little robot and the big robot would move the same way. So I thought, well, I wonder if I can just learn to make it walk like this just by moving my fingers. So I spent hours and hours practicing and could never get it. It just fell over faster than I could react to it. I could make it stand and balance with deep concentration as hard as balancing a ruler on your finger or something. You've got to be really on top of it.

Yeah, so the manual control wasn't going to work, so I went back to working on the full software control for it.

Jessica: Can you explain though, just for the listeners, about what a dynamically balancing biped robot means?

Trevor: Well, there are a lot of ways of building a walking robot, but the way that had been demonstrated before by Honda in particular, the Honda Asimo worked like this, was it had pretty big square feet and the feet were quite stiff, meaning the joint at the ankle was quite stiff. So that you could take this robot and pose it in a pose such that if you drew a perpendicular up from the center of the feet, it would go through the center of gravity, and then with the robot fixed in this pose, you could set it on the ground and take your hands off and it would balance okay. It wouldn't be hard to push it over, but it would be stable. And so that's straightforward to do with computational geometry.

You can calculate where the perpendicular of things goes in three-dimensional space and you know where the center of gravity of the robot is. So that was the way that it worked, that sort of geometrical thing where at every point in time, the perpendicular from the center of the foot that's on the floor always goes through the center of gravity. And that is a particular kind of look to the walk. It's very smooth and stable, but you have to have the legs bent because you can't bounce up and down and it has a particular kind of look to it. But it really can't deal with soft terrain, with walking on grass or anything. So I wanted to build the thing that walks like people do.

Jessica: And in the Honda one, they'd always put a flat surface down, wouldn't they when they did demos of it?

Trevor: They brought their own stage with carefully constructed layers of cross members under the stage, so it had the same stiffness everywhere. I wanted to build something that could just walk anywhere people can. So it had to balance dynamically. And also I wanted it to be able to run and jump and do things like that. The other problem with the stiff jointed robot is you can't jump because each of the joints ... If you think about the knee joint, there's the knee joint and then a gearbox and then a motor. So if you try to land a jump on that thing, there's going to be gear teeth flying everywhere. So I built mine with compressed air power. Used pneumatic cylinders for all the joints.

What's good about them is, well, they're springy and the same amount of force per unit area that muscle produces. About 50 pounds per square inch is what you get from the cross section of human muscle. And indestructible. You could jump and bounce and kick the ground over and over again and it wouldn't do any damage. Because I'd also seen robotics programs, I'd seen some work at Carnegie Mellon where they were building a walking thing closer to the Asimo style, and everyone was terrified to actually do anything with this robot. They had this nice piece of hardware, but they knew that if the software did the wrong thing, it would just break it and then they'd be spending the rest of the month replacing gear teeth.

Carolynn: I just have to ask this question because as you're talking, what's going through my head is all of those YouTube videos we've all seen of Boston Dynamics and they're dogs and they're bipedal and all that. As you're talking, I'm thinking, yeah, I know exactly what he's talking about because we've all seen these videos. When you watch those videos, are you like, "Yeah, I figured that out a long time ago."? What is your emotional reaction to watching?

Trevor: Well, Boston Dynamics is ... I mean, the stuff in the last few years is way ahead of where I got to. It's pretty impressive.

Carolynn: Do you ever feel like, yeah, I figured that out before anyone else was really doing that, or were they simultaneously figuring out-

Trevor: I think they were simultaneously working on something pretty similar, and I think there was a time when I was ahead and then they were ahead and I was ahead and they were ahead.

Carolynn: But were you guys keeping tabs on each other? You knew what they were doing. They knew what you were. Okay.

Trevor: Yeah.

Carolynn: Okay. So it was just a standard mildly competitive situation.

Trevor: Yeah. Well, I mean, at the time they weren't commercial. They were just DARPA funded and were publishing, and I was releasing videos on YouTube, and so it was all ... I don't know. Not super open. Neither of us open sourced our software, but we were happy to demonstrate what we had working.

Carolynn: Well, then my other question for you would be, did you have a commercial use case in mind while you were doing all this tinkering or you're like, "Hey, I'm just going to figure it out because such an interesting thing for me to do and I'll figure out the business case later."?

Trevor: No, I didn't have a very clear idea of what it would be good for. I mean, I thought someday they're going to be robots, and at least some of them ought to be able to walk because they need to go outside and so on. But no, there wasn't really a commercial case for it. And at some point I realized I need more money. I got to find someone to sponsor this. And so I started talking to ... Well, I started trying to see who would buy it. So I mean, the military wants walking robots. I didn't want to work with the military. Combination of vague pacifism and allergy to bureaucracy. For the military, the use case for walking robots is if you're operating in a theater where you can't tell who the soldiers are versus civilians, the only way to know for sure is to send a soldier in and see who shoots at them. And so if you can do that with a walking robot that looks enough like a soldier that people will shoot at it, then you can tell who the bad guys are.

Jessica: Well, when I met you Trevor ... And by the way, you play a very important role in my life because when I came to that party where I met Paul for the first time, you were the person that I talked to the longest at the party, and you had just been featured for your robot for Dexter Pratt fall, I think, on CNN. You were interviewed on CNN and had a little clip, and at the party you had come in from California just to go to the party, and you had whipped out your computer and you're like, "Let me show you the robot." And I remember it was so fascinating. It was carrying a cup of coffee and not spilling it. Do you remember this Trevor?

Trevor: Okay. Yeah, I remember that era.

Jessica: So what point was that in Dexter's lifecycle?

Trevor: Well, that was quite a bit later.

Jessica: Okay.

Trevor: Yeah. So I eventually got it kind of walking. Not real well. Not well enough that you'd be comfortable having it walking on a sidewalk in a crowd of people or something. But it was balancing and walking and could deal with somewhat non-flat floors and obstacles and so on. And you could shove it a little bit and it would recover. And then I started working on the top half of the robot, on the arms, and I thought I was really foresighted by building the top half with the same bolt pattern so eventually I'd just be able to bolt it onto the legs. But to test out the top half of the robot ... The legs were still kind of unreliable, and it was a bit of a miracle every time it worked properly. And so I built a wheeled base to mount the arms on. And so that worked really well. So we had a machine that was still teleoperated. There was an operator wearing a special suit and a special glove that would follow his movements and make the robot hand do the same thing.

Jessica: The robot hand was Monty, right?

Trevor: Yeah, that's right. There was Dexter, the walking robot, and there was Monty, the upper torso on wheels robot.

Jessica: Which, Carolynn, I don't think you were around at dinners at this time, but actually Dexter and Monty played a very important part of Y Combinator, and we'll get to that in a second about how we started Y Combinator. But the robots, when we used to have these dinners when no one had heard of Y Combinator and we'd convince someone important to come to a dinner, we'd be like, there's nothing to see, right? Because it's a bunch of startups. We'd say, "Come back and see Trevor's robots." And it would take up like an hour. It was so fascinating because they were really cool.

Trevor: It distinguished us from the typical VC firm.

Carolynn: They were kind of terrifying.

Jessica: Yeah. They were a little scary. Unique experience. They were scary. They were so big and metal, and they could definitely-

Carolynn: Very Terminator vibes, for sure.

Trevor: Yeah, we put a hole in one of the walls with a runaway robot.

Jessica: Oh, that office. Well, I want to ask you about this. So you're working on the robots, and do you remember when Paul, in 2005, he must've emailed you to say, we want to start something up where we invest in startups. Do you remember the email trying to convince you to do YC with him? Because he really wanted to work with you and Robert again.

Trevor: Yeah. He phoned me, which was unusual. Usually he'd just send a one sentence email.

Jessica: Wow. He did? He phoned you?

Trevor: Yeah. I remember talking to him out in the parking lot for some reason, and he pitched me this idea that we should be the kind of investors that we wished we'd had. When we started Viaweb, we didn't really know anything about business and certainly didn't know anything about how to get investors. And we talked to some official investors, and they wanted things like monthly cash flow projections, and we had no idea how to even do that. And we eventually found some investors like Paul's art teacher's husband, I guess.

Jessica: Julian.

Carolynn: Julian.

Trevor: First investor. Yeah.

Jessica: Julian was your first investor at Viaweb.

Trevor: He was great. He was helpful. He incorporated us, and he was a lawyer, so he knew how to deal with things. And then we had some other investors who were kind of a nuisance.

But mostly it was such an opaque business we thought, let's just make ... I mean, this is Paul's idea. Let's make easy investment the standard way of getting a company going.

Jessica: So when he called you, what did he say? You won't have to do much, except what? He knew you were working on Anybots.

Trevor: Well, at the time it was, let's just try this for a summer. We'll put out a call for startups on his website. We'll get a bunch of applications, we'll read them. So I think he wanted me to read the applications and see if their technology was going to work.

Jessica: The applications were PDF documents. Do you remember?

Trevor: Yeah. Well, I turned them into PDFs. I think the first version was we gave them a template and they would email it into a special mailbox. And then I wrote some software to reformat it so we weren't just looking at a wall of text and at least had the questions bolded and so on.

Jessica: Carolynn, that's the kind of thing that Trevor used to do. We'd be like, "This doesn't seem to be working efficiently." And Trevor would be like, "Okay, let me just whip something up." And so yeah, that makes sense that you wrote something that turned the individual emails into PDFs. Okay.

Trevor: And we printed them all out and we had 300 or more and I remember sitting there with these huge stacks of them where we were sort of treating them like assignments or something, and we'd write our numerical scores on them.

Jessica: So Paul said, we're going to do this project over the summer, which we called the Summer Founders Program, and you believed in the idea and you thought, okay, this'll be a fun thing to do and get to hang out with Paul and Robert. And of course I was involved too. But do you remember coming for the interviews and stuff in Cambridge?

Trevor: Yeah, that was so exciting. There were all these people that had these good ideas of things to start. And so I remember coming out to Cambridge and we'd interview. At the time, we did half hour interviews, right?

Jessica: Oh yeah. Even longer maybe.

Trevor: We might've had 45 minute slots or something.

Jessica: I think they were 45 minute slots.

Trevor: We never would've known that before we did it. These are the kind of things you only learn by doing it. So we-

Jessica: We learned some of the painful ones.

Trevor: We rarely changed our opinion after the first five minutes. And so let's just keep the interview short.

Jessica: And the ones where you knew this is a no, the next 40 minutes were really painful. We were recently interviewing the Stripe brothers, John and Patrick, and they basically said, "How weird is it that in your very first batch you got the founders of Twitch, ultimately Twitch, the founders of Reddit, Sam Altman." Was that shocking to you that the people we got were so good in that first batch?

Trevor: Well, I guess that's what we were hoping for is we'd get good founders. I didn't have a preconceived idea really for how many good ones we should expect. I think a lot of investors think you have to get nearly 100%, and every one that goes wrong is like, you've got to do a post-mortem and understand how you screwed up. And we knew from the beginning that it wasn't going to be like that. Maybe only half of them would work out.

Jessica: I think we had lower expectations than that, Trevor. I think we were psyched just to get some decent programmers and see what happens. We weren't expecting half of them to succeed. Did you come out that first summer? Did you come out and spend some time at all and attend any of the dinners and meet some of the first founders?

Trevor: Yeah, I think I went to more than half the dinners that first time. It was a bit of a strain traveling back and forth, but it was worth it. The dinners were the most energizing thing in my life at the time because it was these really interesting guys working on cool stuff. And I liked Reddit, the product. I liked many of the products. I liked Kiko, the calendar that Emmett Shear and Justin started that eventually turned into Twitch.

Jessica: Yes, I know.

Carolynn: Long path later.

Trevor: Through a series of pivots.

Jessica: So do you remember then we felt like this is pretty interesting, this summer founders program. Do you remember when Paul and I came out to California and said, "Well, Trevor, we don't want anyone else to be the Y Combinator of Silicon Valley, so we're going to do a three-month program in Silicon Valley. How about we borrow some of your office?"

Trevor: Yeah. How hard could it be to turn this low-ceilinged robot lab into a startup dinner theater?

Jessica: You were very accommodating. I mean, you did have a big warehouse kind of thing. You had a lot of space, right?

Trevor: Yeah, I had a lot of space. It had been this warehouse for a religious book shipping company. And so there's this big warehouse, but then the office part of it was pretty drab with those low acoustic ceilings and stuff.

Jessica: Green carpet.

Trevor: Green.

Jessica: That still lives in one room. That was all over the place, Carolynn. The whole place was the green carpet and the low hung acoustic ceilings. And the fluorescent lights.

Trevor: Yeah.

Jessica: So you said, okay. And we said, we just need a little space. And then-

Trevor: Well, and then Kate got to work on designing it because she designed a lot of the Cambridge office.

Jessica: So Kate Corto, who was an architect by trade, oversaw the illegal renovation of part of Trevor's office space.

Trevor: As far as I know, it was completely legal. At some point, the city disagreed and shut it down.

Jessica: Did the city find out?

Trevor: Yeah. Well, I think you were still in Cambridge or something, but the city found out that we were doing this pretty major renovation on the inside. As I understand it, they usually get tips from the people taking away the dumpsters. They see all this construction debris and they don't see a construction permit.

Jessica: Oh my God. Okay, keep going.

Trevor: So they shut us down and put this big red stop work order on the front door. So we couldn't get any of the official contractors in. A union electrician or something wouldn't come. Kate got some of her less formal friend contractors to keep working, but we couldn't get the electrical work done. So I actually wired up a lot of the lights in order to ... For that first dinner, we had people coming on the first Tuesday night, and they shut down the work just a week or so before, and we didn't have the lights on.

Jessica: Oh my god.

Trevor: So I was up on one of those skyjack things, wiring up the lights.

Jessica: I'm in denial that this trauma had happened. I'm now remembering this. So Kate had gotten pretty far though, Trevor, right? And we'd ripped out the hung ceilings, ripped out a lot of the carpet, but a few weeks before the city got wind, busted the construction, and then that's when we started locking the door. And some people still came in and painted. You did the electrical work. Is that what happened?

Trevor: Yeah.

Jessica: And that's why we were so nervous. Because we'd already been busted, right?

Trevor: Yeah. Right. And we were probably pushing the limits on our fire occupancy permit too for some of those things.

Carolynn: I feel like I want to unhear a lot of. Just kidding.

Jessica: Okay. Maybe let's change the subject. And so the paint was drying on the walls.

Trevor: Yeah. We had to tell people not to lean on the walls that first night.

Jessica: Yeah, because the paint was still drying. Oh my goodness. And that's when part of the YC dinner experience was to come to the back of the room and see these robots of Trevor's, and it was really part of the whole charm of early YC.

Trevor: Yeah.

Jessica: Wow. Do you remember any other crazy stories from the first batch in Mountain View?

Carolynn: Yeah. What are your favorite anecdotes from those early days?

Trevor: I remember the Twitch founders. That started off as something called Justin TV, which was basically Justin Khan had a camera strapped to his head and a giant backpack full of computer stuff. Because at the time, you couldn't just stream video over the internet from a phone or anything. You had to have this complicated system. So they had this backpack with four different cell phone antennas in it that Kyle Vogt had built. And then Kyle went on to found Cruise, the self-driving car company.

So I remember them bringing this ... They had this backpack full of stuff and something went wrong with it and so they were using my lab bench and soldering iron and so on to try to get it running again. And then this was in the very early days of streaming when people thought that calling the police on a streamer would be extremely funny. And well, the first time they did it, they just ordered a bunch of pizzas for Justin.

Jessica: I remember. Wasn't it during a YC dinner?

Trevor: It was during a dinner, yeah. So Justin was there at the dinner streaming somehow, and they thought, ha-ha. We'll order these pizzas and we'll get to see him be all confused by the pizza delivery. But actually everyone was hungry and the pizzas-

Carolynn: Someone else bought you all pizzas. No downside.

Jessica: I was very annoyed because the pizza delivery guys showed up during the dinner. We had dinner already and we had a speaker coming and this pizza delivery guy's like, "Okay, I need payment." Of course, the pranksters didn't pay.

Carolynn: Oh, I thought the pranksters paid.

Jessica: The guy had 30 pizzas and expected me to pay, which I did because I was like, it's Justin TV again. There was always stuff like that happening. There were definitely some crazy stories I feel like that happened within those walls. The toilets always broke down. We only had two toilets for everyone, and that was fine when we only had 10 startups, but-

Trevor: Yeah. This thing was basically like an office for 20 people plus a warehouse. So it was at its limits when we were having 100 founders. And then when we did the demo days, we'd have 300 investors show up and all try to park their BMWs on this little side street.

Jessica: This little cul-de-sac.

Trevor: And there'd be one bathroom, and we eventually got porta-potties out in the parking lot for them.

Carolynn: Is that true?

Jessica: Which are not really that classy.

Carolynn: Which was a little below the standard they're used to.

Well, does people-

Jessica: Nowhere to wash your hands.

Carolynn: What about the secret third bathroom in the very back? Is that the sneaky bathroom? Was that just for Anybots people at the time? You didn't let founders back there?

Trevor: Yeah, right. Because I was also trying to run a robot company in that space.

Jessica: I was very protective of the sneaky bathroom, but by the way, I say it was a sneaky bathroom. At first. We finally redid it, of course, illegally. But it was really gross. It was like-

Trevor: Yeah, it was like the warehouse bathroom.

Jessica: Yeah. It was the warehouse bathroom. It was really dirty with old cans of stuff.

Trevor: That's where we'd go pour out paints or something.

Jessica: Yeah, there were paint cans and stuff. And I remember that Ashton Kutcher, who's an excellent investor by the way, and he was coming to our demo days very early on. He was married to Demi Moore at the time, and he brought her to one of the demo days. And I remember meeting her and the line at demo day of the two bathrooms in front, which were the "nice bathrooms", was out the door. And I remember saying, "Well, I do have a special bathroom in the back." I was like, "But it's really gross." And God bless her, she went to this nasty bathroom in the back with garbage and stuff.

Carolynn: Paint cans.

Jessica: I was so embarrassed. I'm like, Demi Moore just used the sneaky bathroom and I'm so embarrassed.

Carolynn: That's funny.

Trevor: There's motivation to clean it up a little.

Jessica: Yeah. But there were definitely some funny stories in that place.

Carolynn: So Paul told a story about you and note cards, and I feel like we have to get your side of that story. So do you want to talk about the note cards?

Jessica: We must.

Trevor: I don't think it's nearly as eccentric as Paul portrays it, but at the time I had this idea, and it actually came from John McPhee, the author. There's a book about his process. Maybe it was just a New Yorker article, how he writes books. And so he said he'd go do all the research and put it on note cards and then spread all these note cards out on the floor like this whole four of his office until he felt like he had everything in the order that he wanted to write about it in and then he'd just go through it and write it. So I thought I should do this for my thesis. I'll just make notes on note cards of every idea. I'll take my class notes on them even. And at some point I'll find these two cards from disparate classes and there'll be some great idea that comes from putting them next to each other.

Jessica: Did that ever happen? The great idea?

Trevor: Nope. No. It was actually a terrible system. It was bad for taking notes because actually a note card, like a five by seven note card is a little too small for a whole idea of information, especially for a math class or something with a lot of formulas and derivations on it.

Carolynn: As you've described it, I agree with you. Not eccentric. Unusual but not eccentric. But I thought that Paul was making it sound like these were note cards you would bring to parties and you would use them as discussion topics when you were just meeting people and chatting with them. It was like they were like ... So did you ever do that?

Jessica: Let me pull out my family card. Let me pull out my card on this topic. Did you ever do that? We need to know.

Trevor: No, I don't think I ever did that.

Carolynn: I wonder what Paul's thinking of.

Jessica: Okay. What about not at a party but in the office? Were you ever like, "Hold on while I pull out this note card."?

Trevor: Yeah, I'm sure I did that.

Jessica: Did you ever consult the note cards? Basically?

Trevor: Yeah. Yeah.

Carolynn: Okay. Well, if all they had on them were interesting academic ideas versus cocktail conversation ideas, then I agree it's less eccentric than I was led to believe in the Paul Graham story.

Jessica: Or what about administrative data? Didn't you have a note card on your car?

Trevor: Yeah, I had to-dos on them.

Carolynn: Oh, that's just the checklist manifesto right there.

Trevor: Yeah. No, checklists are good.

Carolynn: Checklists are good.

Jessica: That didn't last long, the whole card thing?

Trevor: No, no.

Jessica: Trevor, speaking of being eccentric, do you remember when you were on the Segway polo team and Woz was on your team? Can you tell us a little bit about that?

Trevor: Yeah. When the Segway came out, I thought it was an interesting idea. It kind of looked really dorky, but it's a new kind of transportation. No one had done that design before. It's different from a bicycle or something. And I had this fascination with balance from walking robots, so I thought I know how to make one of those. So I made one, and in a couple of weeks I had something that worked and it was pretty clunky. It had a little potentiometer for steering, so it had these big handles for driving, but this little knob for steering. I wrote about that on the internet and that got a bunch of hits on my website and I got invited to this Segway polo group. There's this group of Silicon Valley nerds, including Steve Wozniak who would play polo on Segways. So they had polo mallets and a ball and would drive around on the Segways. And it sounds ... Well, it's about as nerdy as it sounds I guess, but it was surprisingly athletic actually.

Carolynn: I actually believe that.

Trevor: Polo on horseback is athletic too. You've got to be pretty strong.

Carolynn: That I believe.

Jessica: But on the Segway?

Trevor: Yeah, because you're leaning and crashing.

Jessica: Were you playing on your Segwell, which you reverse engineered or did you have your Segway?

Trevor: Right. I went to one of these things and brought my home-built one, which-

Jessica: Excellent.

Trevor: Had some advantages. It could spin around really fast. The Segway, you turn the thing and it goes ... And turns around. Takes like five seconds to do a rotation. Where mine would just go ...

Carolynn: And fling you off in the process.

Trevor: Right. Well, I didn't put any limit on it in the software. I didn't know what the limit should be, so I just didn't put one. So it would go as fast as the motors could rotate. And so yeah, I took it to a couple of these games and eventually they said, first of all, it's lots of exposed metal, whereas a proper Segway is all sort of pad, slightly soft plastic around it and it's way faster.

Carolynn: Wait, they kicked you off the team because you had a technological advantage. So instead of just wanting you on their team, they were like, "Yeah, this guy can't play at all. Forget it. Drop him." That is not cool.

Jessica: Yeah, they should have respected the fact that you built this thing.

Carolynn: Why didn't they put in orders for you to build them their machines? Well, actually we should ask Trevor what he's doing now. So Anybots eventually ... What happened to Anybots actually?

Trevor: I just finished packing it up and put it in a shipping container and shipped it to England. And so the robots are coming.

Carolynn: They're coming.

Trevor: Supposed to arrive-

Carolynn: They're going to live in your barn.

Trevor: Any week now.

Jessica: Oh, I can't wait to see them.

Carolynn: Long lost friends.

Trevor: Yeah.

Carolynn: That's actually sad.

Jessica: But you're still working on stuff.

Trevor: Just software for robots. Yeah. I'm trying to write software that will sort of learn to control the body better. There have been all these advances in machine learning in the last few years and some of them have been applied to robots, but not very many. So that's kind of what I'm working on, is trying to figure out how to apply this machine learning stuff, which now works to controlling robots.

Jessica: Well, if there's anyone who can do that, I bet it's you.

Carolynn: Yeah. I'm glad you're still working on them. I didn't know that that was still happening, so I think that's awesome.

Trevor: Yeah. Well maybe I'll figure it out.

Jessica: Awesome, Trevor. I had so much fun catching up with you and getting your perspective because often I'll tell the story of YC and I kind of know what Paul thinks about it too, but I have never really asked you what you thought about some of the stuff that happened in the early days. And of course hearing about Monty and Dexter, the robots. And I'm glad you're still working on robot related stuff because it's just amazing. And I think this is going to be a really fun episode when it launches.

Carolynn: It's like a full circle.

Trevor: Yeah.

Jessica: So thank you for coming on the show and it's great catching up with you.

Trevor: Yeah, you too. Thanks for having me.

Carolynn: Thanks Trevor. See you.

Jessica: All right, bye.

Trevor: Bye.

Jessica: Oh, Carolynn, that was so much fun talking to Trevor.

Carolynn: Yeah, it's great to talk to him. And I think he cleared up the note card story, although Paul's version is funny.

Jessica: I know.

Carolynn: But I actually thought that Segway polo is pretty funny. It's a pretty funny story.

Jessica: Segway polo. I'm so glad he played on that nerdy Segway polo team because the whole reason I was able to get Steve Wozniak to be interviewed for Founders at Work was through Trevor's connection.

Carolynn: Oh, no kidding.

Jessica: That was my connection. I didn't know Steve Wozniak.

Carolynn: Oh, yeah. Yeah.

Jessica: That was very critical. And also, I mean, not to make a joke out of it, but when I met Paul at his party, like I said, Trevor was the person I talked to the longest and all about his robot.

Carolynn: That's serendipity. What if he hadn't shut up at that party?

Jessica: I know. I might've left out the back door thinking-

Carolynn: Nobody here I want to talk to. Right, right, right. That was very lucky.

Jessica: One thing that didn't come up was just how much fun he is to work with. And in the interviews I'll never forget ... I mean we have some long days with the interviews, as you know.

Carolynn: I do.

Jessica: And you get kind of punch drunk. And Trevor had this knack of every time saying something that would make me go into fits of giggles and just completely lose it. Makes me laugh so hard. And he's also saved my bacon several times because he's the person you call if your computer crashes and you can't even log on or you can't get on. He's resurrected my laptop before.

Carolynn: Oh, that's so nice.

Jessica: He just can get stuff done like that.

Carolynn: That's awesome.

Jessica: And I'm glad to learn a little bit about his robots. Because they were really important at the time and he never got any publicity really for it.

Carolynn: And actually-

Jessica: Because he didn't try.

Carolynn: I was just thinking I'm happy that he gets his robot family back to him in England, but I'm sad they're not going to be in our space anymore because they're definitely a part of the YC folklore from my perspective since I saw them also, the first time I came to see you guys. I got a robot tour also. So we've all been through it and the fact that they're not going to be there anymore is a little bit sad, but it's okay.

Jessica: All right. Well, you'll just have to come visit us in England, Carolynn.

Carolynn: No problem. I will do that.

Jessica: All right. This was a great episode and I can't wait for it to come out.

Carolynn: That sounds great. See you soon.

Jessica: See you soon.

Carolynn: Bye.

Jessica: Bye.